Technical Note

Recency Bias Is Architecture: Axis-First Ordering as a Structural Requirement for Context-Governed LLM Sessions

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March 2026 | CC0 1.0 Universal

Abstract

Recency bias in transformer-based language models — the tendency to over-weight recently presented context relative to earlier context — is commonly attributed to model capability and addressed through prompt engineering. This note presents empirical evidence that recency bias is an attention architecture property, not a capability limitation: it persists identically in Qwen2.5-14B and Claude Sonnet 4 across three independent experimental runs at full STP-derived budget. The note further demonstrates that (1) conservative retrieval system prompts reliably worsen performance by lowering the reporting threshold for early-context information, and (2) axis-first ordering — placing the session origin unit first in the context window — completely resolves recency bias without any selection governance. The STP structural minimum formula (W = 500 + depth×200 + switches×100) is validated as the correct budget oracle: budgets below the structural floor produce deterministic failures regardless of governance policy. These findings establish two invariants for any context-governed LLM session: the window must be sized by structural counters, and the axis must be positioned first.

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1. Introduction

Context-governed LLM sessions declare an axis at t=0 — a hash-committed origin representing the session's ground truth — and measure all subsequent content against that origin. The omega-session system (Mohan, March 2026) implements this governance via structural counting: nesting depth, context switches, and derived window demand. The STP structural minimum formula determines the budget required to hold the full argument structure of the ingested corpus.

An open question in the architecture concerns what happens when the governed window is correctly sized but the axis unit is not positionally privileged. Standard context assembly builds the window in recency order — most recent turns closest to the model's response generation. If recency bias causes the model to over-weight recent context, the axis unit at the session origin is the most vulnerable piece of information in the window regardless of budget.

This note reports Test T8: a controlled experiment feeding a structured philosophical essay through four context assembly conditions across two models, measuring recall of a planted

thesis claim present only in the axis unit. The experiment was designed to answer three questions:

- Is recency bias a capability limitation (solvable by a more capable model) or an attention architecture property (model-independent)?
- Do conservative retrieval system prompts help or harm recall of early-context information?
- Is axis-first ordering a sufficient fix, or does selection governance remain necessary?

2. Method

2.1 Corpus

The test corpus is a 1,109-token philosophical essay on the Computational Theory of Mind, structured around Putnam's (1967) Multiple Realizability thesis. The essay was segmented using the truth sentinel document segmenter (parse_context_units), producing five content units after filtering heading-only stubs:

Unit	sw_idx	~Tokens	Opening text	Role
0	0	302	The Multiple Realizability thesis, first formalized…	AXIS
1	2	111	However, Searle's Chinese Room argument (1980)…	PIVOT
2	3	202	Yet Block's access-phenomenal distinction…	PIVOT
3	4	238	## Information-Theoretic Accounts Tononi's…	PIVOT
4	6	256	Nevertheless, the functionalist core survives…	PIVOT

2.2 Planted Fact and STP Structural Minimum Budget

Planted fact: Lewis, Armstrong, and Shoemaker as the canonical formalizers of functionalism. This triple appears only in Unit 0. No pivot unit contains all three names.

STP structural minimum: W = 500 + (0×200) + (6×100) = 1,100 tokens. Stack depth = 0, context switches = 6. This is the structural floor derived from the essay's own argument structure — the minimum budget required to hold all five units.

Budget zones:

- budget < 302 tokens — INVALID: axis unit cannot fit; all conditions fail by construction
- 302–1,100 tokens — TIGHT: axis fits but argument structure partially lost
- ≥ 1,100 tokens — VALID: full structure preserved

2.3 Conditions

Four conditions were tested for each model:

Condition	Assembly method	Purpose
A guided	Recency order + system prompt: 'only report from context'	Baseline with explicit retrieval instruction
A blind	Recency order, no system prompt	Pure model behavior, no instruction
A' (axis-first)	Same units as A, but unit 0 placed first regardless of recency	Tests whether ordering alone resolves recency bias
C (depth-stratified)	Units selected by switch_index priority; axis always kept first	Structural governance — selection + placement

The recall probe was: "Going back to the opening section: which three specific philosophers are named as providing the canonical formalization of functionalism, and what is the central claim about pain that they all share?"

Recall was scored as the count of {Lewis, Armstrong, Shoemaker} present in the response. all_three requires all three names.

2.4 Models

- Qwen2.5-14B via Ollama (local, Apple M3 32GB) — Conditions A, B (lexical axis), C
- Claude Sonnet 4 (claude-sonnet-4-20250514) via Anthropic API — Conditions A guided, A blind, A', C

Sonnet conditions were run three times at STP-derived budget to verify stability.

3. Results

3.1 Qwen2.5-14B

Condition	Score	Budget mode	Note
A (sliding)	0/3	STP-derived (1100)	Confabulates Putnam/Fodor — 5 turns kept
B (lexical axis)	0/3	STP-derived (1100)	73% eviction; drops Armstrong/Shoemaker
C (depth-stratified)	3/3 ✓	STP-derived (1100)	Unit 0 protected; 20% eviction
A (sliding)	0/3	Stress 400 (36%)	2 turns kept — axis evicted
B (lexical axis)	3/3 ✓	Stress 400 (36%)	88% eviction; unit 0 kept (Ω=0 by construction)
C (depth-stratified)	3/3 ✓	Stress 400 (36%)	1 unit kept (axis only); 80% eviction

3.2 Claude Sonnet 4 (3 runs at STP-derived budget, 1 run at stress 400)

Condition	Run 1	Run 2	Run 3	Stress 400	Note
A guided	0/3	0/3	0/3	0/3	Stable fail — backfire
A blind	2/3	2/3	2/3	0/3	Stable partial at STP budget; evicted at stress
A' (axis-first)	3/3	3/3	3/3	3/3	Stable pass — ordering fix confirmed
C (depth-stratified)	3/3	3/3	3/3	3/3	Stable pass — structural governance

3.3 The Serial Position Effect

Condition A blind consistently scores 2/3, finding Lewis and Armstrong but not Shoemaker across all three STP-budget runs. Shoemaker appears in Unit 0 as the third item in a three-name list: "Lewis, Armstrong, and Shoemaker." Under recency-ordered context, Unit 0 receives the lowest attention weight. Within Unit 0, the third list item receives the lowest positional weight. The consistent dropout of the final list item is not stochastic — it reflects the interaction of recency bias with serial position effects within a single unit.

This predicts a testable asymmetry: if the planted triple were listed as "Shoemaker, Armstrong, and Lewis," the blind condition would consistently drop Lewis, not Shoemaker.

4. Discussion

4.1 Finding 1: Recency Bias Is Architecture, Not Capability

Qwen2.5-14B and Claude Sonnet 4 produce identical Condition A scores at both budget levels. Sonnet is substantially more capable at instruction following and context reading than Qwen — this is established across standard benchmarks. The equivalence in Condition A failure cannot be attributed to capability differences.

The failure mode differs in surface presentation — Qwen confabulates Putnam/Fodor/Lewis from training knowledge; Sonnet correctly reports uncertainty ("I cannot find an opening section...") — but the structural outcome is identical: zero recall of the planted fact under recency ordering. The information was present. The budget was structurally valid. The model had everything it needed. The failure is positional, not informational.

This establishes that recency bias is a property of transformer attention weighting that persists across model scale. It cannot be addressed by deploying a more capable model.

4.2 Finding 2: System Prompt Backfire

Condition A guided scores 0/3 across all runs. Condition A blind scores 2/3 at STP-derived budget. The system prompt — "you ONLY report what was explicitly stated in the conversation above" — consistently degrades performance relative to the uninstructed baseline.

The mechanism: conservative retrieval instructions lower the reporting threshold. Under recency bias, Unit 0 already receives reduced attention weight. The conservative instruction converts the model's reduced-confidence reading of Unit 0 into a zero-confidence report: "I do not see it" rather than "I see it faintly." Sonnet's instruction-following capability works against the user when the instruction's conservatism interacts with architectural attention weighting.

Operational implication: explicit retrieval instructions are not reliably helpful for early-context information. The correct intervention is structural placement, not instruction.

4.3 Finding 3: Axis-First Ordering Is the Minimum Viable Fix

Condition A' passes at all budget levels, across all runs, for both models. The only difference from Condition A is that Unit 0 is placed first in the context window. Same model, same budget, same units, same system prompt in the guided variant. One placement change produces a deterministic transition from 0/3 to 3/3.

This establishes axis-first ordering as a structural invariant for context-governed sessions. The axis unit must be positioned first in the context window — not merely present, not merely protected from eviction, but positionally first. This is a placement rule, not a selection rule. It costs nothing to implement and resolves Condition A completely at adequate budget.

4.4 The Role of Selection Governance Under Budget Pressure

At stress 400 (36% of STP structural minimum), Condition A' still passes because the axis unit fits within the 400-token budget (302 tokens) and is placed first. Selection governance (Condition C) also passes. Both work at stress 400.

Selection governance becomes necessary when the budget falls below the axis unit size (below 302 tokens for this corpus). At that point, no ordering fix helps — the unit cannot fit. The STP structural minimum formula identifies this floor precisely: it is the budget level at which structural information loss becomes deterministic regardless of policy. All conditions failed identically at stress 280, which is below the 302-token axis unit size.

The two-layer model is therefore:

- Axis-first ordering: resolves recency bias at adequate budget. Necessary always.
- Selection governance (C / CDP): resolves budget pressure when window is tight. Necessary when budget < STP structural minimum.
- STP structural minimum formula: determines which regime applies. Computed from structural counters before any budget decision.

5. The STP Structural Minimum as Budget Oracle

The STP structural minimum formula was introduced in the security primitive paper (Mohan, March 2026) as the threshold at which structural counting triggers governance events. T8 validates it in a different role: as the minimum context budget required to hold a corpus's full argument structure. Earlier work referred to this as the 'IEP minimum window' — the canonical name is STP structural minimum, reflecting the March 2026 standardisation that unified STP

and IEP as the same kernel at three application scopes: session, turn, and compaction boundary.

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W = STP_BASE + (depth × c₁) + (switches × c₂)      [STP structural minimum]
W = 500 + (0 × 200) + (6 × 100) = 1,100 tokens    [T8 corpus, depth=0, switches=6]
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For the T8 corpus: depth=0, switches=6, giving W=1,100. The actual corpus token count is 1,109 — the formula predicts the structural minimum to within 1%. This is not coincidence: the formula encodes the relationship between argument complexity and the context required to hold it.

The budget oracle principle: do not set a context window budget by guessing or by percentage of corpus. Run parse_context_units() first, compute W, then size the window. A budget below W guarantees structural information loss. A budget below the axis unit size guarantees total failure. Both failures are diagnosable in advance from the structural counters alone.

One measurement, three functions: security gate threshold (structural counting triggers NACK above this level), context window budget oracle, and epistemic cost budget (Conservation invariant I2). The same formula addresses all three because adversarial complexity, argument complexity, and context demand are the same structural property measured from different angles. This is why the formula is a primitive property of the STP kernel, not of any specific application scope.

6. Relationship to the Omega-Session Architecture

The omega-session paper (Mohan, March 2026) demonstrated that governed eviction outperforms sliding window eviction for technical scientific corpora (T5, T7) and philosophical corpora (T8) under budget pressure. T8 adds two architectural requirements not present in the original specification:

- Axis-first placement: the axis unit must be first in the assembled context window, not merely retained through eviction policy.
- STP structural minimum validation: the window must be sized at or above the STP structural minimum before any eviction policy is applied.

These requirements are independent of the eviction policy used. A system implementing CDP (directional Ω vectors with per-switch local axes) still fails Condition A if the assembled context is recency-ordered. CDP governs selection; axis-first governs placement. Both are required.

The complete two-layer specification for a context-governed session:

Layer	Mechanism	Invariant
Budget sizing	STP structural minimum from counters	budget ≥ W = 500 + depth×200 + switches×100
Placement	Axis-first ordering in context assembly	Unit 0 is always first in the window
Selection (if budget < W)	Depth-stratified or CDP governance	Axis protected; pivot units deprioritised

6.1 The Three Failure Regimes

T8 tested Regimes 1 and 2. A third regime exists that T8 did not test, identified from the observation that context windows are periodically compacted in production sessions — reducing early context to keyword summaries. This creates a qualitatively different failure mode from recency bias within a live window.

Regime	Condition	Symptom	Fix
1	budget ≥ stp_min, axis present	Model over-weights recent context (recency bias)	Axis-first placement — tested in T8
2	axis_tok ≤ budget < stp_min	Axis present but argument structure lost under eviction	STP structural minimum + depth-stratified selection — tested in T8 stress runs
3	Compaction boundary crossed	Axis compressed to keywords; hash gone; session appears valid but runs against lossy summary	Hash-committed axis stored outside window; reinjected at session re-establishment — not yet tested

Regime 3 is the most dangerous because it is invisible without the hash commitment. The model appears oriented. Responses remain coherent. The axis has simply been replaced by an approximation, and no downstream check can detect this without the original hash. The omega-session architecture addresses Regime 3 through the STP handshake's cryptographic axis commitment — the axis lives outside the compaction boundary and is reinjected identically on session re-establishment. Empirical validation of Regime 3 is the next experimental target.

6.2 The Ego-Dissolution Extension: Perspective Depth Pass

The depth-stratified governance in Condition C uses switch_index as a proxy for axis proximity — units before the first switch are foundational, units after each switch are pivots or counterarguments. This is correct for hard lateral switches but misclassifies a structurally important class: perspective descent and return pairs.

In the philosophical essay, 'However, Searle argues...' is not a lateral switch — it is a perspective entry: you step into Searle's frame, the prior frame (Putnam/Lewis/Armstrong) goes onto the stack. 'Nevertheless, the functionalist core survives...' is the return: you ascend back to the parent frame. The switch_index scorer assigns both the same tier (pivot, switch_index > 0). The perspective depth pass distinguishes them:

- PUSH — perspective entry (However, Searle argues...) — depth increases, moving away from axis
- POP — perspective return (Nevertheless, functionalism survives...) — depth decreases, returning to axis
- HOLD — continuation within current frame — no depth change

Unit	sw_idx	op	depth	ω dir	priority	Text
0	0	HOLD	0	−0.3	(0, 0.0)	The Multiple Realizability thesis... [AXIS]
1	2	PUSH	1	+0.7	(4, −1.0)	However, Searle's Chinese Room...
2	3	PUSH	2	+0.7	(4, −1.0)	Yet Block's access-phenomenal...
3	4	HOLD	2	+0.3	(3, 1.0)	## Information-Theoretic Accounts...
4	6	POP	1	−0.7	(2, −1.0)	Nevertheless, functionalist core survives... [RETURN]

Unit 4 (Nevertheless/POP) has omega_direction = −0.7 — it is returning toward the axis. The switch_index scorer (Condition C) ranks it as a low-priority pivot (sw=6). The perspective depth scorer (Condition D) promotes it to priority tier 2 — just below the axis itself — because it is axis-convergent.

Empirical Result at Stress 600

At budget 600 tokens (55% of STP structural minimum):

- C (switch_index): kept [0, 1] — axis + Searle descent (PUSH, moving away from axis)
- D (perspective): kept [0, 4] — axis + Nevertheless return (POP, returning to axis)
- Both conditions recalled all three philosophers correctly

The structural choice is architecturally correct. C kept a descent unit. D kept a return unit. The empirical separation between C and D requires a corpus where the return unit carries information the axis unit does not — where the thesis is introduced briefly and elaborated in the conclusion after counterarguments. In the current corpus, the axis unit alone contains the planted fact, so both conditions succeed. This is a corpus design property, not a failure of the perspective depth pass.

Lexical Scorer Degradation at Stress 450

At budget 450 tokens (41% of STP structural minimum), the lexical axis scorer (Condition B) failed (0/3) while both structural position scorers (C and D) passed (3/3). The lexical scorer has no structural protection — under sufficient budget pressure it evicts the wrong units regardless of axis commitment. Structural position scorers protect the axis by construction, independent of vocabulary signal.

Budget	A sliding	B lexical	C switch	D perspective	Note
1100 (STP min)	0/3 ✗	3/3 ✓	3/3 ✓	3/3 ✓	All structural pass
600 (55%)	0/3 ✗	3/3 ✓	3/3 [0,1]	3/3 [0,4]	D keeps return unit
450 (41%)	0/3 ✗	0/3 ✗	3/3 ✓	3/3 ✓	B degrades, struct holds
420 (38%)	0/3 ✗	3/3 ✓	3/3 ✓	3/3 ✓	B recovers

Grammar as Axis — Future Direction

The current perspective depth pass uses pattern matching on English discourse markers. The Internal Ear Protocol (Mohan, January 2026) establishes the correct foundation: natural language grammar encodes perspective shifts structurally. Every language has a subject/agent marking system, a subordination mechanism, and a discourse cohesion system. A dependency parse extracts perspective operations directly from grammatical structure rather than from language-specific marker lists.

The immediate path: replace the regex pattern tables with spaCy dependency parse subject extraction. When the ROOT subject changes and the transition is contrastive, that is a PUSH. When the ROOT subject returns to the axis subject after a nested perspective, that is a POP. This generalises across all languages spaCy supports — the grammar is the transition function, the axis subject is the Cpigment, and the Markovian succession $S_{n+1} = f(S_n, C_p^{IGMGRH}) + \Delta$ operates on grammatical subject continuity rather than lexical markers.

7. Falsifiable Predictions

Prediction 1 — Serial position reversal: If the planted triple is listed in reverse order (Shoemaker, Armstrong, Lewis), Condition A blind will drop Lewis rather than Shoemaker across multiple runs. The consistent dropout is the last item in the list, not a specific philosopher.

Prediction 2 — System prompt threshold: There exists a conservative retrieval system prompt aggressive enough to cause Condition A blind to score 0/3. The backfire is not binary; it scales with instruction conservatism.

Prediction 3 — A' robustness: Condition A' passes for any model, at any budget above the axis unit token count, for any corpus, when the axis unit contains the recall target. The axis-first fix is not corpus-specific.

Prediction 4 — STP floor universality: The STP structural minimum predicts budget failure deterministically: at budget < axis_unit_tokens, all conditions fail regardless of model, ordering, or governance policy. No experiment will find a condition that passes below this floor.

8. Conclusion

Intelligence is context, correctly positioned.

The T8 experiment establishes that recency bias in transformer attention is an architectural property, not a capability limitation. State-of-the-art models fail identically to smaller models when context is recency-ordered, and the failure is stable and deterministic. Conservative retrieval instructions reliably make this worse, not better.

The fix is structural and cheap: place the axis unit first. This single placement rule deterministically resolves Condition A failure at adequate budget, across all models tested, in every run. No selection governance, no system prompt engineering, no model upgrade required.

The STP structural minimum tells you the required budget before you run. The axis-first rule tells you how to assemble the window once you have the budget. Together they are sufficient for reliable axis recall at the STP structural minimum level. Selection governance (CDP)

becomes necessary only under budget pressure — and the formula tells you exactly when you are under pressure.

The context window is the solvency meter. The axis goes first. The formula tells you how large the meter must be.

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